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Device, kit and method for correction of spinal deformity

Abstract

A device, kit and method for the correction of spinal deformity using an anterior approach. The device comprises a screw-head platform, wherein the screw-head platform comprises a primary recess for receiving a primary rod, and a main vertebral body screw rotatably connected to the screw-head platform.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation application of U.S. patent application Ser. No. 16/539,011, filed Aug. 13, 2019, which is a continuation application of U.S. patent application Ser. No. 13/937,460, filed Jul. 9, 2013, which claims the benefit of priority to U.S. Provisional Patent Application Ser. No. 61/670,903, filed Jul. 12, 2012, entitled "A DEVICE FOR CORRECTION OF SPINAL DEFORMITY," the teachings of each of which are hereby incorporated by reference as if set forth in their entirety herein.

TECHNICAL FIELD

(1) The present relates to device, kit and methods for correction of spinal deformity. The device is particularly useful in the area of anterior thoracolumbar deformity surgery, including the treatment of scoliosis and deformity related to trauma, infection, tumour and degeneration.

BACKGROUND

(2) The instrumented correction of spinal deformity, and particularly scoliosis, using an anterior approach has been proven to be a successful technical concept. Clinical and scientific work has shown it to be an ideal technique for treating the majority of idiopathic scoliosis types.

(3) In comparison to posterior scoliosis surgery, the use of very early anterior derotation and fusion for selective treatment of thoracic and thoracolumbar curves was shown to save up to two fusion levels compared to the posterior approach. Further, the anterior approach does not disrupt the posterior muscles, it is better at restoring sagittal profile of the spine, it enables real derotation through an anterior column release and direct vertebral manipulation close to the centre of rotation.

(4) As noted above, the use of modern instruments to provide an anterior scoliosis fixation provides significant advantages. A known approach is the use of an instrument with two rods. The use of such a double rod system improves the ease of reconstructing and maintaining a physiologic sagittal spino-pelvic alignment. An exemplary two rod system is described in U.S. Pat. No. 5,702,395. This document describes a spinal osteosynthesis instrumentation for an anterior approach. The instrumentation comprises two rigid rods parallel to each other, and at least two blocks adapted to transversely interconnect the rods, as well as bone anchorage elements for the blocks. The two blocks and the two associated rods make a rectangular frame, which is said to afford all of the required stability for restoring the axial continuity of the vertebral column, avoids correction losses, and facilitates the consolidation of the bone graft.

(5) Recent developments in implant technology have increased the biomechanical stabilisation potential of anterior constructs. Hence, the number of pathologies that can be managed with anterior-only strategies has rapidly increased.

(6) However, with both anterior-only scoliosis correction and reconstruction of the anterior column in other deformities, the amount of corrective force applied to the instrumentation is limited by the interface strength and pull out resistance of the screws anchoring the instrument to the vertebral

body. Hence, further development of implants and screws used for anterior spinal surgery has the potential to increase construct stability, broaden the indications for which anterior column surgery is suitable and push the limits of current techniques and corrections that are currently achieved.

SUMMARY

(7) It is an object of the present disclosure to improve the angular stability of a device for correction of spinal deformity using an anterior approach.

(8) It has been found that connecting a main vertebral body screw to a screw-head platform so that the main vertebral body screw cannot move in its axial direction relative to the screw-head platform advantageously ensures that the main vertebral body screw has a stable angular relationship relative to the screw-head platform. This angular stability of the main vertebral body screw relative to the screw-head platform ensures that the screw direction is at a constant angle relative to the screw-head platform. Therefore, there is limited possibility for potentially unpredictable angular changes between the main vertebral body screw and the screw-head platform during or after implantation. Such unpredictable changes may be particularly problematic after implantation as it will adjust how the spine is being corrected. Further, the connection of the main vertebral body screw to the screw-head platform reduces the number of parts that need to be assembled during surgery and thus decreases the complexity of the implantation procedure. It also reduces the possibility of losing either of the main vertebral body screw or screw-head platform during the operation as they are connected to each other.

(9) It has also been found that providing a screw-head platform that aligns a main vertebral body screw in the same plane as an anchor screw reduces the required size of the screw-head platform, in particular, in the cranial-caudal direction. This is therefore a more efficient use of space inside the vertebral body, in particular, thoracic and cervical vertebral bodies, to which the device will be affixed, where space is limited. Further, the triangulation of the screws increases the resistance against pull out and cyclic loads. Such an arrangement can also be used to ensure that the anchor screw, which is attached after the main vertebral body screw, does not extend beyond the main vertebral body screw and potentially damage the spinal cord. This can be achieved by making sure that the path of the anchor screw converges to a point along the main vertebral body screw and so would contact the main vertebral body screw before penetrating beyond it.

(10) According to the present disclosure, there is provided a device for correction of spinal deformity using an anterior approach comprising a screw-head platform, wherein the screw-head platform comprises at least one or a primary recess for receiving a primary rod; and a main vertebral body screw rotatably connected to the screw-head platform.

(11) The present disclosure also provides a device for correction of spinal deformity using an anterior approach comprising a screw-head platform, wherein the screw-head platform comprises at least one recess, for instance a primary recess, for receiving a primary rod. The screw-head platform comprises a main bore for receiving a main vertebral body screw and an anchor bore for receiving an anchor screw. The main bore and the anchor bore are positioned so as to align the anchor screw at an oblique angle relative to the main vertebral body screw in a common plane with the main vertebral body screw.

(12) As stated above, the devices, kits, and methods described herein are suitable for use with any anterior thoracolumbar deformity surgery. These include the treatment of scoliosis and deformity related to trauma, infection, tumor and degeneration.

(13) The screw-head platform is the component through which the one or more corrective rods (primary rod and, possibly, secondary rod) are attached to the patient's spine in order to correct the spinal deformity. The screw-head platform is attached to the vertebral body of the patient by the main vertebral body screw. In alternative embodiment, the device includes an anchor screw. In general, it is desirable to have a screw-head platform that is as small as possible while still possessing its full function and mechanical stability. In particular, a low profile of the screw-head platform above the vertebral surface to which it is attached is very important due to the limited

space around the patient's spine. These space constraints are particularly acute in the case of correcting spinal deformities in younger patients. The screw-head platform can have a low profile and a low volume.

(14) The underside of the platform is shaped in a complementary manner to the surface of the vertebra to which it is to be attached. In this way, the height of the screw-head platform off of the vertebral surface can be minimized. The underside of the screw-head platform can be concave, in particular, in a longitudinal direction, or cranial-caudal direction when implanted, to complement the anatomical shape of the surface of the vertebral body. The underside of the screw-head platform refers to the screw-head platform surface that faces the vertebra when the device is in use. The underside can be the surface through which the main vertebral body screw and anchor screws protrude to affix the screw-head platform to the vertebra.

(15) The screw-head platform comprises at least one, for instance a primary recess, that is configured to receive a primary rod. The primary recess can define a U-shaped channel running through the screw-head platform with an open top. The open U-shape of the channel allows the primary rod to be easily inserted into the recess. The recess can have a width that corresponds approximately to the diameter of the primary rod. The width of the recess can allow the rod to be inserted into the recess while reducing the lateral (i.e. along a lateral direction that is perpendicular to the length direction of the rod) movement of the rod within the recess. This decreased potential for movement increases the overall mechanical stability of the deformity correction system.

(16) Securement of the primary rod can be further supported by the primary recess having a complementary shape to that of the primary rod. This can increase the size of the contact area between the screw-head platform and the primary rod and so further reduces the potential movement, particularly lateral movement, of the primary rod when it is fully seated in the primary recess. The primary rod may also be held or contacted within the recess by a component that is shaped in a complementary manner to the primary rod so as to restrict the primary rod's movement, in particular its lateral movement. Such an additional contacting component may be part of the securing means described below. The configuration of the primary recess applies to the secondary recess, described below, if present.

(17) The screw-head platform is generally elongate in shape along the lateral direction. The primary recess for receiving the primary rod generally runs through the screw-head platform in a direction that is perpendicular to the screw-head platform's elongate direction or lateral direction. The elongate shape of the screw-head platform can accommodate the main vertebral body screw and an anchor screw.

(18) The main bore in the screw-head platform is sized to receive the main vertebral body screw. The main bore may be shaped to have a complementary surface to the underside of the head of the main vertebral body screw so that the main vertebral body screw is supported in its axial direction when the main vertebral body screw is fully seated within the main bore.

(19) The main bore is positioned at the base of the primary recess so that the main bore, and the head of the main vertebral body screw when it is in the main bore, may be accessed via the recess prior to the insertion of the primary rod into the primary recess. After insertion of the primary rod in the primary recess, the primary rod obstructs access to the main bore or the head of the main screw in the primary recess. This arrangement keeps the size of the screw-head platform small as independent access to the main bore, distinct from that of the primary recess, does not have to be accommodated for in the screw-head platform.

(20) As noted above, the main vertebral body screw may be rotatably connected to the screw-head platform. This has the effect of restricting the axial movement of the main vertebral body screw relative to the screw-head platform, while still allowing relative rotational movement. In other words, the rotatable connection ensures that the vertebral body screw cannot move in the axial direction of the main vertebral body screw, i.e. the movement of the main vertebral body screw in its axial direction is substantially prevented. The axial direction of the main vertebral body screw

runs along the length of the screw and is the direction in which the screw penetrates the body to which it is being affixed. The inability of the main vertebral body screw to move in its axial direction has the results highlighted above. Namely, the angular relationship between the main vertebral body screw and the screw-head platform will remain constant as restraining the main vertebral body screw from moving in the main vertebral body screw's axial direction eliminates any differential axial movement across the diameter of the main vertebral body screw. Such differential axial movement would cause the alignment of the main vertebral body screw with the screw-head platform to change. Further stability is achieved by the axial movement restriction by ensuring that the main vertebral body screw is retained properly seated within the main bore of the screw-head platform. This means that the head of the main vertebral body screw can be fully supported by the complementary surface of the main bore.

(21) The main vertebral body screw axial movement restriction removes the angular relationship between the main vertebral body screw and the screw-head platform as a concern during surgery and so the use of such a device increases the speed and reliability of performing a spinal deformity correction. Further, the axial retention of the main vertebral body screw also minimises the possibility that the main vertebral body screw will move relative to the screw-head platform during subsequent manipulation of the vertebra to which it is connected. This again increases the reliability of the device and the predictability of its subsequent behaviour after being affixed to the patient's vertebra. Also, connecting component parts of the device together before surgery reduces the chance of losing either component during the operation.

(22) The connection between the main vertebral body screw and the screw-head platform is such that it restricts relative axial movement but allows relative rotational movement, i.e. it fixes the axial movement of the main vertebral body screw relative to the screw-head platform but it does not fix the rotation movement of the main vertebral body screw relative to the screw-head platform. This allows the main vertebral body screw to be screwed into the vertebra body of the patient while maintaining the screw-head platform in the desired rotational orientation. The screw-head platform may comprise a retention component to perform the function of connecting the screw-head platform to the main vertebral body screw, restricting relative axial movement while allowing relative rotational movement. Such a retention component may be an integral part of the screw-head platform or it may be a separate component that is attached to the screw-head platform.

(23) In order to provide the connection between the main vertebral body screw and the screw-head platform, the main vertebral body screw may have an annular indentation. The screw-head platform may then have a retention component that can interact with this annular indentation in order to constrain the axial movement of the main vertebral body screw. In particular, the retention component of the screw-head platform may protrude into the annular indentation provided on the main vertebral body screw and so constrain its axial movement. It will be appreciated that a protrusion that closely fits the annular indentation would minimize the play between the protrusion in the annular indentation.

(24) The retention component may comprise an annular protrusion so that the retention component protrudes into the annular indentation of the main vertebral body screw all the way around the screw. However, the retention component may only protrude into part of the annular indentation. For example, the retention component may protrude into the annular indentation so that it protrudes into between 180° to 360° of the annular indentation. An example of such a retention component would be a C-clip. Alternatively, the retention component may only protrude into less than 180° of the annular indentation of the main vertebral body screw, alternatively less than 90° of the annular indentation of the main vertebral body screw. The retention component may comprise a single continuous protrusion that is received in the annular indentation of the main vertebral body screw. Alternatively, the retention component may comprise a plurality of discrete protrusions that protrude into the annular indentation. A higher amount of the retention component protruding into the annular indentation of the main vertebral body screw will improve the stability of the axial

constraint. Decreasing the proportion of the annular indentation that contains a protrusion of the retention component will lower the frictional resistance provided by the retention component to the main vertebral body screw's rotational movement.

(25) Other retention components are of course possible as the skilled person would understand. For example, the retention component could comprise a pin arranged to engage with the annular indentation. In one example, a pin or a plurality of pins may be passed through the screw-head platform into the annular indentation, the pins then being fixed in position. In another example, the pin or pins may be biased to protrude from the screw-head platform, the pin, or pins, being displaceable against this bias. This allows the main vertebral body screw to be inserted into the main bore of the screw-head platform by displacing the pin, or pins, out of their protruding arrangement. The pin, or pins, will then resiliently snap back to protrude into the annular indentation to restrict further axial movement of the main vertebral body screw.

(26) Where the retention component comprises a plurality of discrete protrusions that protrude into the annular indentation, the plurality of discrete protrusions can be arranged symmetrically when protruding into the annular indentation of the main vertebral body screw. This reduces the presence of less stable directions of the main vertebral body screw. Further, the plurality of discrete protrusions number at least three to ensure a certain level of stability is provided by the retention component.

(27) As indicated above, the main vertebral body screw is able to rotate relative to the screw-head platform in order to allow it to be screwed into the vertebra of the patient. The above described retention components are such as to allow this relative rotational movement while inhibiting the axial movement of the main vertebral body screw.

(28) The screw-head platform may further comprise an anchor bore. The anchor bore is sized so as to receive an anchor screw. The anchor bore may be such that when it receives the anchor screw, it aligns the anchor screw at an oblique angle relative to the main vertebral body screw. In other words, the anchor bore is such so as to align the anchor screw so that the line along the length of the anchor screw (the anchor screw's axial direction) is angled with respect to the line running along the length of the main vertebral body screw (the main vertebral body screw's axial direction) so that the anchor screw's axial direction is neither parallel nor at a right angle to the main vertebral body screw's axial direction. The presence of an anchor screw in addition to the main vertebral body screw improves the pull out resistance of the device. This is of great benefit because conditions such as scoliosis are often associated with poor bone quality of the vertebra.

(29) The anchor screw may have angular stability relative to the screw-head platform. This angular stability increases the overall reliability and stability of the device. Such angular stability can be provided by the anchor bore comprising a thread that can interact with a thread on the head of the anchor screw as the anchor screw is secured into the anchor bore.

(30) The anchor screw is not connected to the screw-head platform in the manner described above with respect to the main vertebral body screw above. While anchor screw is configured so that axial movement of the anchor screw is possible. In this way, the anchor screw will not necessarily be present when securing the main vertebral body screw into the patient's vertebra, which could result in the anchor screw interfering with this operation. This thus avoids an increase in the complexity of the securing operation.

(31) The anchor bore may also be positioned so as to align the anchor screw in a common plane with the main vertebral body screw. Hence, the anchor screw's axial direction and the main vertebral body screw's axial direction will both fully lie in a same plane. Such an arrangement allows the screw-head platform to remain small resulting in efficient use of space within the patient's body and within the vertebral body to which the screw-head platform is attached. Such an arrangement of the anchor screw and the main vertebral body screw is also beneficial in terms of biomechanical concerns, increasing the stability of the device.

(32) The angle between the anchor screw's axial direction and the main vertebral body screw's axial

direction when both screws are properly seated in the screw-head platform is based on the anatomical situation. In particular, it has been found that the range of angles of 15° to 30° is particularly acceptable. In an exemplary embodiment, the angle θ is approximately 20° degrees between the anchor screw's axial direction and the main vertebral body screw's axial direction. The angle between the anchor screw's axial direction and the main vertebral body screw's axial direction is the acute angle formed between the axial direction of the main vertebral body screw and the axial direction of the anchor screw. The axial directions of the screws run along the length of the screws and in the direction in which the screw will travel when being screwed into a body.

(33) The common plane between the anchor screw and the main vertebral body screw is perpendicular to the direction in which the primary rod runs when received in the primary recess of the screw-head platform, i.e. perpendicular to the length of the primary rod when it is received in the primary recess. Again, this ensures that the screw-head platform can maintain a small dimension in the direction parallel to the primary rod's length when received in the primary recess. The reduction of the dimensions of the screw-head platform has the advantages highlighted above.

(34) The height of the screw-head platform is approximately 15 mm or less, for example, the height of the screw-head platform may be 12 mm or may be 10 mm. The height of the screw-head platform is the distance measured perpendicular from the screw-head platform surface that faces the vertebra body in use to the surface opposite. In other words, the screw-head platform height is a measure of how far the screw-head platform rises above the underside of the screw-head platform. Ensuring that the screw-head platform's height is as low as possible improves the space efficiency of the device and reduces the interference caused by the screw-head platform in situ.

(35) The screw-head platform is of a generally rectangular shape. Specifically, the underside of the screw-head platform, which contacts the vertebra body, is of a rectangular shape to accommodate the screws and the rods. Both the length and the width of the rectangular screw-head platform are 30 mm or less, in particular, the length and width of the rectangular screw-head platform are 25 mm or less. Again, this keeps the screw-head platform compact so that it does not interfere when in situ and efficiently uses the space available in the patient's body.

(36) As noted above, the screw-head platform is shaped to receive a primary rod. The screw-head platform may also comprise a secondary recess, similar to the primary recess, for receiving a secondary rod. This secondary recess runs parallel to the primary recess in order to receive a secondary rod that is parallel to a primary rod. The secondary recess is sized in accordance with the size of the secondary rod that is to be used. The use of two rods results in a supportive rectangular frame that helps support the readjusted spine.

(37) The device of the present disclosure may comprise a means for securing the primary rod to the screw-head platform. The device may also include a means for securing the secondary rod to the screw-head platform.

(38) The means for securing the primary rod or the means for securing the secondary rod may comprise a top loading screw. A top loading screw seals the recess into which the rod is placed. In other words, the rod is placed in the open recess and the top loading screw secures the rod by blocking the top of the recess so that an enclosed channel is formed in which the rod is held. The rod is inhibited from rotating or sliding within the channel by the top loading screw applying a sufficient pressure to the rod. A top loading screw as a securing means can be used for just one of the primary recess or secondary recess. Alternatively, a top loading screw can be used as a securing means with both the primary recess and the secondary recess.

(39) The use of a top loading screw is advantageous as it provides an easier way of securing a rod. This is due to the relatively wide tolerance in the position of the rod prior to the commencement of securing using the top loading screw. Screwing of the top loading screw then gradually confines the rod to its secure position within the recess. Top loading screws also reduce the length and/or width requirements of the screw-head platform as they secure the rod by aligning with a central axis of the rod. Whereas, the use of side loading screws, considered below, requires additional room within

the screw-head platform in which to accommodate a screw besides the received rod.

(40) An alternative means for securing the primary rod or for securing the secondary rod is a side loading screw. Side loading screws are positioned adjacent the recess and possess a conical head. The conical head overhangs the recess. Therefore, loosening the side loading screw increases the gap into which the rod can be placed. After placing of the rod in the recess beneath the conical head of the side loading screw, the screw is tightened so that the rod is held in position by the conical head. Such side loading screws occupy less height for a given mechanical stability when compared to top loading screws. However, it can be difficult to introduce a rod into a recess with a side loading screw and subsequently secure the rod.

(41) Although any combination of top loading and side loading screws can be used with the device of the present disclosure, a top loading screw can be used with the primary recess as previous work has shown that using the top loading screw is easier to attach to the screw-head platform. The secondary rod can be secured by either a side loading screw or a top loading screw, which will result in the relevant advantages described above. It is possible to use a side loading screw for securing both the primary rod and the secondary rod. However, the use of two side loading screws would make the platform longer, which could be problematic in certain spinal deformity surgery such as with adolescent vertebrae, where space within the patient is restricted. In particular, it is possible have a top loading screw for securing the primary rod and a top loading screw for securing the secondary rod. Alternatively, it is possible to have a top loading screw for securing the primary rod and a side loading screw for securing the secondary rod.

(42) The device may comprise a primary rod for securing in the primary recess. The device may also include a secondary rod for securing in the secondary recess. The primary rod can have a diameter of 5.0 mm or greater. Further, the primary rod can have a diameter of 5.5 mm or less. In an exemplary embodiment, the primary rod is about 5.5 mm in diameter. The secondary rod has a diameter of 3.5 mm or greater. Further, the diameter of the secondary rod is 4.0 mm or less.

(43) The device may also comprise an anchor screw. The anchor screw is received in the anchor bore as described above.

(44) All of the parts of the device may be made from any suitable biocompatible material. Exemplary materials are metallic materials. Suitable metallic materials include cobalt-chrome alloys, titanium, and stainless steel.

(45) The device comprises a plurality of screw-head platforms each of the screw-head platforms receives and secures the primary rod, and possibly the secondary rod, to various parts of the spine. The combination of the platforms and the rod(s) as described herein allows the spine to be manipulated and supported for correcting spinal deformities.

(46) Each of the plurality of screw-head platforms can have the features described above. In particular, each of the screw-head platforms may comprise the same features or different features to each of the other screw-head platforms. In particular, each of the plurality of screw-head platforms will have a main vertebral body screw associated with them so that each platform can be secured to the patient's spine, offering support and allowing manipulation.

(47) In general, when in use the screw-head platforms are secured to the lateral circumference of the vertebra body and the underside of the screw-head platform rests on the surface of the vertebra. The screw-head platform is shaped with a curvature that enables the underside of the screw-head platform to fit best to the vertebra body. Such curvature will minimise the gap between the underside of the screw-head platform and the body of the vertebra. This also reduces the profile of the screw-head platform off of the vertebra body.

(48) The present disclosure also provides a kit for correction of spinal deformity using an anterior approach. The kit includes a screw-head platform, wherein the screw-head platform comprises a primary recess for receiving a primary rod and a main bore for receiving a main vertebral body screw; a main vertebral body screw; and a retention component for connecting the screw-head platform to the main vertebral body screw so that the main vertebral body screw cannot move

relative to the screw-head platform in the axial direction of the main vertebral body screw.

(49) Such a kit of parts may comprise additional features or parts as recited above. The kit of parts for assembling a device of the present disclosure enables the device to be in a disassembled state for ease of transport and to be then subsequently assembled before or during the operation in which it will be used.

(50) The retention component referred to above may be integral to the screw-head platform or it may be a separate component which is subsequently secured to the screw-head platform so as to perform the task of retaining the main vertebral body screw relative to the screw-head platform.

(51) The present disclosure also provides a method of correcting a spinal deformity using an anterior approach with the devices of the present disclosure, comprising the steps of affixing a screw-head platform to a patient's spine by screwing the main vertebral body screw into the lateral surface of the vertebral body; inserting the primary rod into the screw-head platform's primary recess; and securing the primary rod within the primary recess.

(52) The use of a device of the present disclosure with the above described method improves the ease with which the surgery can be performed as well as the reliability of the procedure. This is due to the various features described above. For example, connecting the main vertebral body screw to the screw-head platform decreases the likelihood that the screw will be lost during the operation as well as reducing the chance that the angle between the main vertebral body screw and the screw-head platform will change during or after the operation. Further, the presence of an anchor screw that is aligned by the screw-head platform in a common plane with the main vertebral body screw reduces the required size of the screw-head platform and so decreases the space required when implanting the screw-head platform in the patient's body.

(53) In addition to the screw-head platform being affixed by the main vertebral body screw, an anchor screw can also be screwed into the lateral surface of the vertebral body, through the screw-head platform, so as to further secure the screw-head platform in position. The anchor screw can be screwed into the vertebral body after the main vertebral body screw has been screwed into position.

(54) After inserting the primary rod into the primary recess, the primary rod may be secured within the primary recess by applying and screwing a top loading screw so as to contain the primary rod between the top loading screw and the screw-head platform.

(55) The method may comprise affixing a plurality of screw-head platforms to the patient's spine and then inserting a primary rod into each of the screw-head platforms and securing the primary rod in each of these screw-head platforms. The insertion of the primary rod into each of the screw-head platforms may be associated with a level of spinal manipulation. The securing of the primary rod then holds this manipulation in place. In this way, the patient's spine can be manipulated and supported to correct the spinal deformity.

(56) The method may also comprise inserting a secondary rod into a secondary recess in the screw-head platform. This is then followed by the securing of the secondary rod within the secondary recess, either by applying and screwing a top loading screw or by screwing a side loading screw so as to contain the secondary rod between the screw-head platform and the screw. The securing of the secondary rod occurs after the securing of the primary rod, the primary rod being typically designed to withstand greater forces than the secondary rod. When there are a plurality of screw-head platforms present, the primary rod can be secured in all of the plurality of screw-head platforms before the secondary rod is secured in any of them. Alternatively, the primary rod and then the secondary rod can be secured into each screw-head platform, primary rod first, before commencing the insertion and securing of the primary rod and secondary rod in the next screw-head platform.

(57) The device of the present disclosure can be implanted by performing a thoracotomy, a thoracolumbophrenotomy, or by using an endoscopic approach. The vertebral column is exposed and the lateral surface of the vertebral body is dissected to remove the soft tissue. A small awl is used to form a pilot hole for the main vertebral body screw. This pilot hole is usually close to the

plane of the rib heads, parallel to the end plates in the axial plane, albeit perpendicular to the lateral vertebral body surface slightly directing anteriorly off the spinal canal. The length of the screw and screw tract preparation can be assessed based on previous MRI and/or CT based measurements of vertebral depth, using the surgeon's finger approaching the contralateral vertebral body wall or by assessment of vertebral depth at the discectomy level adjacent to the vertebra to be instrumented. The screw platform is then inserted in the desired orientation and screwed into place using a screw driver with the main vertebral body screw. Then the second smaller anchor screw is secured to the vertebra. Finally, the rods are locked to the screw-head platform. If there are multiple rods of different sizes, the larger diameter rod is secured first and then the smaller diameter rod.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The foregoing summary, as well as the following detailed description of illustrative embodiments of the device of the present application, will be better understood when read in conjunction with the appended drawings. For the purposes of illustrating the device of the present application, there is shown in the drawings illustrative embodiments. It should be understood, however, that the application is not limited to the precise arrangements and instrumentalities shown. In the drawings:
- (2) FIG. 1 is a cross-sectional view of a device for correcting a spinal deformity taken along lines 1-1 in FIG. 2, according to a first embodiment of the present disclosure;
- (3) FIG. 2 is a perspective view of a device according to the first embodiment of the present disclosure;
- (4) FIG. 3 is a cross-sectional view of the device illustrated in FIG. 4, taken along line 3-3 thereof, according to a second embodiment of the present disclosure.
- (5) FIG. 4 is a perspective view of a device shown in FIG. 3, according to the second embodiment of the present disclosure;
- (6) FIG. 5 is a sectional view of a device according to a third embodiment of the device of the present disclosure; and
- (7) FIG. 6 illustrates a device according to third embodiment of present disclosure shown implanted in situ in the patient's body.

DETAILED DESCRIPTION OF THE DRAWINGS

- (8) Certain terminology is used in the following description for convenience only and is not limiting. The words “right”, “left”, “lower” and “upper” designate directions in the drawings to which reference is made. The words “inner” or “distal” and “outer” or “proximal” refer to directions toward and away from, respectively, the geometric center of the implant and related parts thereof. The words, “anterior”, “posterior”, “superior,” “inferior,” “medial,” “lateral,” and related words and/or phrases are used to designate various positions and orientations in the human body to which reference is made and are not meant to be limiting. The terminology includes the above-listed words, derivatives thereof and words of similar import.
- (9) Referring to FIGS. 1 and 2, the device **100** includes a platform member **102**, referred to herein as a screw-head platform **102**, a first anchor or main vertebral body screw **104**, and an second anchor **106** or anchor screw **106**. The platform member **102** can define one or more recesses configured to receive at least one elongate corrective rod **114**. The main vertebral body screw **104** is rotatably connected to the platform member **102** while also being configured to secure to a vertebral body (not shown). The main vertebral body screw **104** can secure the platform member **102** to the vertebral body. In addition, the second anchor **106** can be inserted through the platform member **102** into the vertebral body. The platform member **102** is configured such that the anchor screw **106** is aligned at an oblique angle relative to the main vertebral body screw **104** and is

disposed in common plane **107** with the main vertebral body screw **104**, as further detailed below. The screw-head platform **102** can attach one or more corrective elongate rods, for instance a primary rod **114** (FIGS. 1-2) and a secondary rod **224** (FIGS. 3-6), to the patient's spine in order to correct the spinal deformity. Further, the screw-head platform **102** is configured to have a low-profile and low volume. A screw-head platform **102** with a low profile or distance relative to the vertebra and a low volume can accommodate the limited spaced constraints around the spine as noted above. Further, the main vertebral body screw **104** can be connected to the screw-head platform **102** a retention component **108**. The retention component **108**, which in the drawing is illustrated as a C-clip, protrudes into an indentation **109** defined by the main vertebral body screw **104**. The retention component **108** resists movement of the main vertebral body screw **104** relative to the screw-head platform **102** along the vertical direction V as further detailed below. The device also includes a locking means, for instance a top-loading screw **110** or threaded plug **110** that can secure the primary rod **114** in the primary recess **112**.

(10) Referring to FIGS. 1 and 2, the first anchor **104** and the second anchor **106** can be configured to engage with the screw-head platform and the vertebral body. The first and second anchors **104** and **106** can be any device configured to anchor or secure to a bone. For instance, the first anchor **104** includes a head **126** and a shaft **127** that extends from the head **122** along a screw axis **104a**. Further, the second anchor includes a head **118** and shaft **119** that extends from the head along a second screw axis **106a**. The screw axes **104a** and **106a** can extend along, and are aligned with, the axial directions AD1 and AD2 of respective anchors. Further, the first and second anchors can be entirely threaded (FIGS. 3 and 5), or partially threaded (FIG. 1) along an outer surface thereof. The phrase “first anchor” is used interchangeably with the phrase “main vertebral body screw” and the phrase the “second anchor” is used interchangeably with the phrase “anchor screw” throughout the present disclosure.

(11) Continuing with FIGS. 1 and 2, the platform **102** is configured to position the first anchor **104** relative to the second anchor **106**, and also to attach one or more corrective rods to the vertebrae of a spine. The platform **102** defines upper surface **133** and a lower surface or underside **132** spaced from the upper surface **133** along the vertical direction V. The platform member **102** defines opposed sides **134** and **136** that are spaced apart along the lateral direction A, and opposed ends **138** and **140** (FIG. 2) spaced apart along the longitudinal direction L. The device **100** is described herein as extending along a longitudinal direction “L”, a lateral direction “A”, and vertically along a vertical or transverse direction “V”. Unless otherwise specified herein, the terms “longitudinal,” “lateral,” and “vertical” are used to describe the orthogonal directional components of various device components and device component axes. It should be appreciated that while the longitudinal and lateral directions are illustrated as extending along a horizontal plane, and that the vertical direction is illustrated as extending along a vertical plane, the planes that encompass the various directions may differ during use. For instance, when the device is implanted, the longitudinal direction L extends generally along the caudal-cranial direction. The horizontal plane defined by the lateral direction A and vertical direction T lies generally in the anatomical plane defined by the anterior-posterior direction, and the medial-lateral direction.

(12) Referring again to FIGS. 1 and 2, the platform **102** is configured such that the main vertebral body screw **104** is prevented from moving along the vertical direction V relative to the screw-head platform **102**, which provides improved angular stability as discussed above. Further, a screw-head platform **102** that aligns a main vertebral body screw **104** in the same plane as an anchor screw **106**, for instance along the lateral direction A, reduces the required size of the screw-head platform **102**, in particular, in the longitudinal direction L (or the cranial-caudal direction) when implanted. The result is a more efficient use of space inside the vertebral body, in particular, thoracic and cervical vertebral bodies, to which the device will be affixed, where space is limited. Further, the triangulation of the screws increases the resistance against pull out and cyclic loads.

(13) As can be seen in FIGS. 1 and 2, the screw-head platform **102** defines a primary recess **112** for

receiving the primary elongate rod **114**. The primary recess **112** can be in the form of a U-shaped channel running through the screw-head platform **102** along the longitudinal direction L. For instance, the platform **102** can define the recess wall **115** that extends from the upper surface **133** the vertical direction V toward underside **132**. The recess **112** can have an open top or upper portion. The platform **102**, and in particular the recess wall **115** includes first and second recess sidewalls **113a** and **113b** that are spaced apart from each other along the lateral direction L, and recess base **125** that is spaced from the upper surface **133**. The U-shape recess **112** thus permits the primary rod **114** to be easily inserted into the recess **112**. Further, the recess **112** has a width W that corresponds approximately to the diameter D of the primary rod **114**. The width W can be the dimension that extends between the opposed first and second recess sidewalls **113a** and **113b**. When the width W is about the same as the diameter D of the primary rod **114**, the rod **114** can be inserted into the recess **112** while reducing lateral movement, i.e. along the lateral direction L, of the rod within the recess **112**. Decreased rod movement increases the overall mechanical stability of the device when implanted.

(14) The shape of the recess **112** can assist in securing the rod **114** in place. For instance, shape of the primary recess **112** can be complementary to the shape of the primary rod **114**. Increased contact area between the screw-head platform **102** and the primary rod **114** can also reduce the potential movement, particularly lateral movement along the lateral direction A of the primary rod **114** when the rod **114** is fully seated in the primary recess **112**. The device **100** can an additional securing member, such a component **116** can that is configured to restrict movement, in particular lateral movement, of the primary rod **114** in the recess **112**. The component **116** may be integral with or separate from the locking means described herein. Further, the primary recess as described herein is similar to the secondary recess configured to receive therein the secondary rod.

(15) Referring to FIGS. 1 and 2, the underside **132** of the screw-head platform **102** is configured to face the vertebra when the device is implanted. In an embodiment, the underside **132** of the platform **102** is shaped in a complementary manner to the surface of the vertebra to which it is to be attached. In this way, the height of the screw-head platform **102** can be minimized, for instance the height of the platform **102** off the vertebra when implanted can be minimized. The underside **132** of the screw-head platform **102** can be concave. For instance, the underside can be concave with respect to a first axis (not shown) that is substantially perpendicular to the longitudinal L and vertical V directions. The first and second anchors **104** and **106** protrude through the underside **132** as to affix the screw-head platform **102** to the vertebra.

(16) The screw-head platform **102** is generally elongate in shape along the lateral direction A. The elongate shape of the screw-head platform **102** accommodates the main vertebral body screw **104** and an anchor screw **106**. For instance, the screw-head platform **102** defines a main or first bore **122** for receiving the main vertebral body screw **104**, and an anchor bore or second bore **120** for receiving the anchor screw **106**. The main bore **122** extends along a main bore axis **123**. In the illustrated embodiment, the main bore **122** extends from the base **125** of recess to the underside **132** of the platform **102** along the vertical direction V such the main bore axis **123** and the vertical direction V are aligned. Further, the anchor bore **120** extends along and defines anchor bore axis **121**, wherein the anchor bore axis **121** and the main bore axis **123** define an angle θ that extends therebetween. As noted above, the angle θ is oblique, and is preferably less than 90 degrees. The main bore **122** and the anchor bore **120** are configured to align the anchor screw **106** at an oblique angle relative to the main vertebral body screw **104** and in a common plane **107** with the main vertebral body screw **104** along the lateral direction A. For instance, the main bore axis **123** and the anchor bore axis **121** lie in a common plane **107** that extends along the lateral direction A. When the main vertebral body screw **104** is disposed in the main bore **122**, the main vertebral body screw extends along the main bore axis **123**, and when the anchor screw **106** is disposed in the anchor bore **120**, the anchor screw **106** extends along the anchor bore axis **121**. In such a configuration, the screw axes **104a** and **106b** are coaxial with the main bore axis **123** and the anchor screw axis,

respectively.

(17) The main bore **122** sized to receive the main vertebral body screw **104**. The main bore **122** may be shaped to have a complementary surface to the underside **132** of the head **118** of the main vertebral body screw **104** so that the main vertebral body screw **104** is supported in its axial direction AD1 when the main vertebral body screw **104** is fully seated within the main bore **122**. The main bore **122**, and the head **118** of the main vertebral body screw **104**, may be accessed through the recess **112**. After insertion of the primary rod **114** in the primary recess **112**, the primary rod **114** obstructs access to the main bore **122** or the head **118** of the main screw in the primary recess **112**.

(18) The main vertebral body screw **104** may be rotatably connected to the screw-head platform **102**. This has the effect of restricting the axial movement of the main vertebral body screw **104** relative to the screw-head platform **102**, while still allowing relative rotational movement the screw **104**. As noted above, the screw-head platform **102** may comprise a retention component **108** to perform the function of connecting the screw-head platform **102** to the main vertebral body screw **104**, restricting relative axial movement while allowing relative rotational movement. Such a retention component **108** may be an integral part of the screw-head platform **102** or it may be a separate component **116** that is attached to the screw-head platform **102**. By restricting relative axial movement of the main vertebral body screw **104**, but allowing relative rotational movement thereof, the main vertebral body screw **104** to be screwed into the vertebra body of the patient while maintaining the screw-head platform **102** in the desired rotational orientation. In order to provide the connection between the main vertebral body screw **104** and the screw-head platform **102**, the main vertebral body screw **104** may have an annular indentation **109**. The retention component **108** that can be disposed at least partially with this annular indentation **109** in order to constrain the axial movement of the main vertebral body screw **104**. In particular, the retention component **108** can be carried by the underside **132** of the platform and adjacent to the main bore **122** to protrude into the annular indentation **109** provided on the main vertebral body screw **104** and so constrain its axial movement. It will be appreciated that a protrusion that closely fits within the annular indentation **109** can minimize the play between the retention component **108** and the annular indentation **109**.

(19) Continuing with FIGS. 1 and 2, the retention component **108** referred to above may be integral to the screw-head platform **102** or it may be a separate component **116** which is subsequently secured to the screw-head platform **102** so as to perform the task of retaining the main vertebral body screw **104** relative to the screw-head platform **102**. The retention component **108** may protrudes into the annular indentation **109** of the main vertebral body screw **104** all the way around the screw. However, the retention component **108** may only protrude into part of the annular indentation **109**. For example, the retention component **108** may protrude into the annular indentation **109** so that it protrudes into between 180° to 360° of the annular indentation **109**. An exemplary retention component **108** is a C-clip. Further, the retention component **108** may only protrude into less than 180° of the annular indentation **109** of the main vertebral body screw **104**, as well as less than 90° of the annular indentation **109** of the main vertebral body screw **104**. The retention component **108** may comprise a single continuous protrusion that is received in the annular indentation **109** of the main vertebral body screw **104**. In other embodiments, the retention component **108** may comprise a plurality of discrete protrusions that protrude into the annular indentation **109**. A higher the volume of the retention component **108** that protrudes into the annular indentation **109** of the main vertebral body screw **104**, the more stable the axial constraint of the screw **104**. Decreasing the proportion of the annular indentation **109** that receives a protrusion of the retention component **108** can lower the frictional resistance provided by the retention component **108** and permit greater rotational movement of the main vertebral body screw **104**. Other retention components are of course possible as the skilled person would understand. For example, the retention component **108** could comprise a pin arranged to engage with the annular

indentation **109**. In one example, a pin or a plurality of pins may be passed through the screw-head platform **102** into the annular indentation **109**, the pins then being fixed in position. In another embodiment, the pin or pins may be biased to protrude from the screw-head platform **102**. In such an embodiment, the pin, or pins, are displaceable against the bias. This allows the main vertebral body screw **104** to be inserted into the main bore **122** of the screw-head platform **102** by displacing the pin, or pins, out of their protruding arrangement. The pin can resiliently snap back to protrude into the annular indentation **109** to restrict further axial movement of the main vertebral body screw **104**. Where the retention component **108** comprises a plurality of discrete protrusions that protrude into the annular indentation **109**, the plurality of discrete protrusions are arranged symmetrically when protruding into the annular indentation **109** of the main vertebral body screw **104**. This can reduce the presence of less stable directions of the main vertebral body screw **104**. Further, the plurality of discrete protrusions number at least three to ensure a certain level of stability is provided by the retention component **108**.

(20) As discussed above, the screw-head platform **102** may further comprise an anchor bore **120**. The anchor bore **120** is sized to receive an anchor screw **106**. The anchor bore **120** configured such that when it receives the anchor screw **106** therein, the anchor screw **106** is aligned at an oblique angle relative to the main vertebral body screw **104** as noted above. In other words, the anchor bore **120** configured to align the anchor screw **106** such the anchor screw axis **106a** (that extends along the length of the anchor screw **106** or AD2) is angled with respect to screw axis **104a** of the main vertebral body screw **104** (the main vertebral body screw axial direction AD1). Thus, the axial direction AD2 of the anchor screw **106** is neither parallel nor at a right angle to the axial direction AD1 of the main vertebral body screw **104**. The combination of the anchor screw **106** in addition to the main vertebral body screw **104** improves the pull out resistance of the device **100**.

(21) The anchor screw **106** may have angular stability relative to the screw-head platform **102**, which increases the overall reliability and stability of the device. The anchor bore **120** can include a thread configured to engage with a thread on the head **118** of the anchor screw **106** as the anchor screw **106** is secured into the anchor bore **120**. It should be appreciated that the anchor screw **106** is configured such that the movement of the anchor screw **106** along the anchor bore axis **121** is not prevented in the same manner that axial movement of the main vertebral body screw **104** is limited. In this way, the anchor screw **106** will not necessarily be present when securing the main vertebral body screw **104** into the patient's vertebra, which could result in the anchor screw **106** interfering with this operation.

(22) The angle θ between the main body vertebral screw **104** and the anchor screw **106** when both screws are properly seated in the screw-head platform **102** is based on the anatomical situation. In particular, it has been found that an angle θ of between 15° to 30° is particularly acceptable. In an exemplary embodiment, angle θ can be approximately 20° . It should be appreciated that the angle θ can be defined as the angle between the screw axes **104a** and **106a**, the angle between the main bore axis **123** and the anchor bore axis **121**, or the angle between the axial directions AD1 and AD2.

(23) As discussed above, the platform **102** has compact size. The height of the screw-head platform **102** is approximately 15 mm or less, for example, the height of the screw-head platform **102** may be 12 mm or may be 10 mm. The height of the screw-head platform **102** is the distance measured perpendicular from the screw-head platform **102** surface that faces the vertebral body **434** in use to the surface opposite. In other words, the screw-head platform **102** height is a measure of how far the screw-head platform **102** rises above the underside **132** of the screw-head platform **102**.

Ensuring that the height of the screw-head platform **102** is as low as possible improves the space efficiency of the device and reduces the interference caused by the screw-head platform **102** in situ.

(24) The screw-head platform **102** is of a generally rectangular shape. Specifically, the underside **132** of the screw-head platform **102**, which contacts the vertebra body, is a rectangular shape so as to accommodate the screws and the rods. The length L (not shown) measured between opposed

ends **138** and **140** of platform **102** is no greater than 30 mm. Further, the length L of the platform is no greater than 25 mm. In addition, the width X (not shown) measured between opposed sides **134** and **136** of the screw-head platform **102** is no greater than 30 mm. Further, the width X (not shown) is no greater than 25 mm. or less. Such a size keeps the screw-head platform **102** compact.

(25) As described above, the device **100** may comprise one or more elongate corrective rods or elongate members configured to be received in the primary recess **112** of the platform member **102**. The primary rod **114** can have a diameter that is greater than 5.0 mm. Alternatively, or in addition, the primary rod **114** can have a diameter that is 5.5 mm or less. In an exemplary embodiment, the primary rod **114** can have a diameter of about 5.5 mm. Further, the primary rods **214**, **314**, and **414** of the second, third and fourth embodiments can have similar dimensions. In accordance with alternative embodiments, the device may also include a secondary rod **224**, **324**, **424** (FIGS. 3-6) configured to be received and secured in the secondary recess **226**, **326**, **426** Further, the secondary rod **224**, **324**, **424** can have a diameter of 4.0 mm or less.

(26) The device as described herein may be made from any suitable biocompatible material. For instance, the platform, first and second anchors, and locking means can be formed of any biocompatible material. Exemplary materials are metallic materials. Suitable metallic materials include cobalt-chrome alloys, titanium, and stainless steel.

(27) Referring to FIG. 2, the device can include a plurality of the screw-head platforms **102**, main vertebral screws **104**, and anchor screws **106**. The primary rod **114** is depicted as secured within the plurality of screw-head platforms **102** in the manner in which it is secured when the device is implanted in the body. The screw-head platform **102** is generally elongate in shape in a direction perpendicular to the length of the primary rod **114**, or the lateral direction A, when the rod **114** is secured to the screw-head platform **102**. The common plane **107** in which the main vertebral body screw **104** and the anchor screw **106** lie is also perpendicular to the length of the primary rod **114** when secured in the screw-head platform **102**. In other words, the common plane extends along the lateral direction A and is perpendicular to the longitudinal direction L. In general, when in use the screw-head platforms **102** are secured to the lateral circumference of the vertebra body and the underside **132** of the screw-head platform **102** rests on the surface of the vertebra. The screw-head platform **102** is shaped with a curvature that enables the underside **132** of the screw-head platform **102** to fit best to the vertebra body. Such curvature will minimise the gap between the underside **132** of the screw-head platform **102** and the body of the vertebra. This also reduces the profile of the screw-head platform **102** off of the vertebra body.

(28) The device of the present disclosure may comprise a locking means **110**, **210**, **310** for securing the primary rod **114**, **214**, **314** to the screw-head platform **102**, **202**, **302**. Further, as shown in FIGS. 3-6, the device may also include a locking means **220**, **320** for securing the secondary rod **224** to the screw-head platform **102**.

(29) The locking means for securing the primary rod or the means for securing the secondary rod may comprise a top loading screw. A top loading screw seals the rod into the recess into which the rod is placed. In other words, the rod is placed in the open recess and the top loading screw secures the rod by blocking the top of the recess so that an enclosed channel is formed in which the rod is held therein. The rod is inhibited from rotating or sliding within the channel by the top loading screw applying a sufficient pressure to the rod. A top loading screw as a securing means can be used for just one of the primary recess or secondary recess. Alternatively, a top loading screw can be used as a securing means with both the primary recess and the secondary recess.

(30) For instance, referring to FIGS. 1 and 2, the locking means **110**, which can include a top-loading screw or threaded plug **110**, secures the primary rod **114** in the recess **112**. The top loading screw **110** can be unscrewed and removed in order to open the U shaped the recess **112**, which can allow the primary rod **114** to be removed or inserted into the primary recess **112**. The top loading screw **110** defines a central screw axis **129**. When the top loading screw **110** is secured to the rod **114**, the central screw axis intersects a central rod axis of the rod **114**. In addition, the top loading

screw **110** can be rotatably connected to a rod holding component **116** shaped in a complementary manner to the primary rod **114** in order to aid in securing the primary rod **114** in the primary recess **112**. Component **116** contacts approximately half the cross-sectional circumference of the primary rod **114** and transfers the restraining force of the threaded plug **110** to the primary rod **114**. While a top-loading screw **110** is shown in FIG. 1 for secure the primary rod **114** in place, a side-loading screw can be used as well. The side loading screw is further detailed below. The use of a top loading screw is advantageous as it provides an easier way of securing a rod. This is due to the relatively wide tolerance in the position of the rod prior to the commencement of securing using the top loading screw. Screwing of the top loading screw then gradually confines the rod to its secure position within the recess. Top loading screws also reduce the length and/or width requirements of the screw-head platform as they secure the rod from above. Whereas, the use of side loading screws, further detailed below, can require additional room within the screw-head platform in which to accommodate a screw.

(31) Referring now to FIGS. 3 and 4, the device **200** illustrated in FIGS. 3 and 4 have similar components to the device illustrated in FIGS. 1 and 2 and described above. Accordingly, features and aspects of the first embodiment are interchangeable with the features and aspects of the second embodiment. In accordance with the alternative embodiment, the device includes a secondary recess for receiving a secondary spinal rod **224**. For instance, the device **200** includes a screw-head platform **202**, a first anchor **204** or main vertebral body screw **204** and a second anchor **206** or an anchor screw **206**, as well as a primary rod **214** in the primary recess **212** of the screw-head platform **202**. The main vertebral body screw **204** is connected to the screw-head platform **202** via the retention component **208**. Further the platform **202** includes a main or first anchor bore and second bore or anchor bore that is configured similarly to the main and anchor bore described above with respect to FIGS. 1 and 2. In accordance with the embodiment illustrated in FIGS. 3 and 4, the anchor screw **206** includes a head **218** and shaft **219** that extends from the head **218** along an anchor axis **206b**. The head **218** of the anchor screw **206** includes a threaded outer surface that corresponds to threads on the inside of the anchor bore **220** of the platform **202**. The threaded anchor screw **206** aids in angular stability of the anchor screw **206** with respect to the screw-head platform **202**. The primary rod **214** is secured in the primary recess **212** by the presence of a top loading screw **210**, similar to the top loading screw **110** described above. The platform **202** also defines a cavity, which defines the primary recess **212** for receiving the primary rod **214**. The cavity extends partially between the main vertebral body screw **204** and the top loading screw **210** within which the primary rod **214** can be secured. The width of the cavity corresponds approximately to the diameter of the primary rod **214**.

(32) Referring to FIG. 3 the screw-head platform **202** is shaped to receive a primary rod **214**. The screw-head platform **202** includes a secondary recess **226**, similar to the primary recess **212**, for receiving a secondary rod **224**. The secondary recess **226** runs parallel to the primary recess **212** in order to receive a secondary rod **224** that is parallel to a primary rod **214**. The secondary recess **226** is sized in accordance with the size of the secondary rod **224** that is to be used. The secondary recess **226** is a semi-circular channel that contacts the secondary rod **224** around approximately half of the cross-sectional circumference of the secondary rod **224**. The secondary rod **224** is secured in the secondary recess **226** by the use of a side loading screw **210**. The main vertebral body screw **204** is connected to the screw-head platform **202** in a similar manner to that shown with the device of FIG. 1. Also, the anchor screw **206** and the main vertebral body screw **204** are at an oblique angle to each other and within a common plane as noted above, similar to the embodiment shown in FIGS. 1 and 2. Accordingly, the common plane is perpendicular to the longitudinal direction L of the platform **202** and length of the primary rod **214** and the secondary rod **224**. The underside **232** of the screw-head platform **202** is shaped in a complementary manner to the vertebra surface to which it will be attached.

(33) In accordance with an alternative embodiment, the present disclosure includes a side loading

screw **228**, **328** for securing the primary rod **214** or for securing the secondary rod **224** in respective recesses. As can be seen in FIGS. **3** and **4**, an embodiment of a side loading screw **228** is shown positioned adjacent the second recess **226** and at least partially in a screw bore **233**. The side loading screw **228** includes a conical head **230** and shaft **213** that extends from the head **230** along central screw axis **229**. The side loading screw **210** is attached to the platform **202** such that the central screw axis **229** is offset, or does not intersect, a central elongate axis of the second rod **214**. The shaft **213** is disposed in the bore **233** and the conical head **230** overhangs at least a portion of the recess **226** when the side-loading screw **228** is attached to the platform **202**. The head **230** of the side loading screw **228** at least partially defines the secondary recess **226** or channel through which the secondary rod **224** runs. Tightening the side loading screw can decrease the gap between the head **230** and recess **226** and loosening the side loading screw **228** increases the gap into which the rod **214** can be placed. In other words, unscrewing the side loading screw **228** increases the space into which the secondary rod **224** can be placed and subsequently secured by tightening the side loading screw **228** so that the conical head **230** contacts and secures the secondary rod **224**. After placing of the rod **214** in the recess **212** beneath the conical head **230** of the side loading screw **228**, the screw **228** is tightened so that the rod **214** is held in position by the conical head **230**. Such a side loading screw **228** does not increase the overall height of the device for a given mechanical stability when compared to top loading screws **110** and/or **210**.

(34) Although any combination of top loading and side loading screws can be used with the device of the present disclosure, it has been shown that a top loading screw is easier to use with regard to the securing of the first rod into the screw-head platform **202**. The secondary rod **224** can be secured by either a side loading screw **228** or a top loading screw **210** which will result in the relevant advantages described above. It is possible to use a side loading screw **228** for securing both the primary rod **214** and the secondary rod **224**. However, the use of two side loading screws would make the platform longer, which could be problematic in certain spinal deformity surgery such as with adolescent vertebrae, where space within the patient is restricted. In particular, it is possible to have a top loading screw **210** for securing the primary rod **214** and a top loading screw **210** for securing the secondary rod **224**. Alternatively, it is possible to have a top loading screw **210** for securing the primary rod **214** and a side loading screw **228** for securing the secondary rod **224** as illustrated.

(35) FIG. **4** depicts the device of FIG. **3** in a perspective view with a primary rod **214** and secondary rod **224** secured to the screw-head platform **202** in a manner similar to that in which the device is used within the body for supporting and manipulating a spinal deformity.

(36) FIG. **5** illustrates a device **300** similar to the device **100** and **200** as described above. In accordance with the alternate embodiment shown in FIG. **5**, the device **200** includes a top loading screw **328** for securing the secondary rod **324** to the screw-head platform **302**. The screw-head platform **302** is shaped to receive a primary rod **314**. The screw-head platform **302** may also comprise a secondary recess **326**, similar to the primary recess **312**, for receiving a secondary rod **324**. The secondary recess **326** runs parallel to the primary recess **312** in order to receive a secondary rod **324** that is parallel to a primary rod **314**. The secondary recess **326** is sized in accordance with the size of the secondary rod **324** that is to be used. The secondary recess **326** has a semi-circular channel at its base, which receives and supports the secondary rod **324** by contacting the secondary rod **324** around approximately half of its cross-sectional circumference. The top loading screw **328** contacts the secondary rod **324** to retain it in the semi-circular channel of the secondary recess **326**.

(37) FIG. **6** depicts a device of the present disclosure in place within the patient's body. It can be seen that the screw-head platforms **402** are fixed to the vertebral body **434** and the primary rod **414** and secondary rod **424** are secured to the screw-head platform **402** so as to offer support to the spine and help counteract the spinal deformity. As described above, the device of the present disclosure is implanted by first exposing a lateral surface **435** of the vertebral body **434**. The first

screw-head platform **402** is then affixed to a designated vertebra by initially drilling a pilot hole and then screwing the main vertebral body screw **404** into the pilot hole, through the screw-head platform **402**, and into the vertebral body **434**. On fully tightening the main vertebral body screw **404**, the screw-head platform **402** is positioned to extend from the posterior to the anterior of the vertebra and is held in position by the main vertebral body screw **404**. Next, the anchor screw **406** is screwed into the vertebral body **434** to secure the screw-head platform **402**. Next, further screw-head platforms **402** are affixed to other vertebra in a similar manner as described for the first screw-head platform **402**. The primary rod **414** is then secured into the screw-head platforms **402**, one at a time, by placing the primary rod **414** into the primary recess **412** of the screw-head platform **402** and securing it in place by applying and tightening a top loading screw **410**. The securing of the primary rod **414** into each of the screw-head platforms **402** may require the manipulation of the spine. It is this manipulation that the device of the present disclosure supports. Next, the secondary rod **424** is secured into the screw-head platforms **402**, one at a time, by placing the secondary rod **424** into the secondary recess **426** and tightening a side loading screw **428**. This further supports the manipulated spine. The method has been described with reference to a particular embodiment of the present disclosure but it should be clear that the general features of the method can be utilised with any devices of the present disclosure.

(38) Each of the plurality of screw-head platforms **102**, **202**, **302**, **402** can have the features described above. In particular, each of the screw-head platforms **102** may comprise the same features or different features to each of the other screw-head platforms **102**. In particular, each of the plurality of screw-head platforms **102** have a main vertebral body screw **104** associated with therewith so that each platform can be secured to the patient's spine, offering support and allowing manipulation.

(39) The present disclosure also provides a kit for the correction of spinal deformity. The kit includes a screw-head platform **102**, wherein the screw-head platform **102** comprises a primary recess **112** for receiving a primary rod **114** and a main bore **122** for receiving a main vertebral body screw **104**; a main vertebral body screw **104**; and a retention component **108** for connecting the screw-head platform **102** to the main vertebral body screw **104** so that the main vertebral body screw **104** cannot move relative to the screw-head platform **102** in the axial direction of the main vertebral body screw **104**. Such a kit of parts may comprise additional features or parts as recited above. The kit of parts for assembling a device of the present disclosure enables the device to be in a disassembled state for ease of transport and to be then subsequently assembled before or during the operation in which it will be used.

(40) The present disclosure also provides a method of correcting a spinal deformity using an anterior approach with the devices of the present disclosure. The method includes the step of affixing a screw-head platform to a patient's spine by screwing the main vertebral body screw into the lateral surface of the vertebral body and inserting the primary rod into the screw-head platform's primary recess. The method includes securing the primary rod within the primary recess. In addition, the method can include the step of affixing the anchor screw into the lateral surface of the vertebral body through the screw-head platform so as to further secure the screw-head platform in position. The anchor screw can be screwed into the vertebral body after the main vertebral body screw has been screwed into position. After inserting the primary rod into the primary recess, the primary rod may be secured within the primary recess by applying and screwing a top loading screw to contain the primary rod between the top loading screw and the screw-head platform. The method may also include affixing a plurality of screw-head platforms to the patient's spine and then inserting a primary rod into each of the screw-head platforms and securing the primary rod in each of these screw-head platforms. The insertion of the primary rod into each of the screw-head platforms may be associated with a level of spinal manipulation. The securing of the primary rod then holds this manipulation in place. The patient's spine can be thus be manipulated and supported to correct the spinal deformity. The method may also comprise inserting a secondary rod into a

secondary recess in the screw-head platform. This is then followed by the securing of the secondary rod within the secondary recess, either by applying and screwing a top loading screw or by screwing a side loading screw so as to contain the secondary rod between the screw-head platform and the screw. Securing of the secondary rod occurs after the securing of the primary rod, the primary rod being typically designed to withstand greater forces than the secondary rod. When there are a plurality of screw-head platforms present, the primary rod can be secured in all of the plurality of screw-head platforms before the secondary rod is secured in any of them. Alternatively, the primary rod and then the secondary rod can be secured into each screw-head platform, primary rod first, before commencing the insertion and securing of the primary rod and secondary rod in the next screw-head platform.

(41) The device of the present disclosure can be implanted by performing a thoracotomy, a thoracolumbophrenotomy, or by using an endoscopic approach. For instance, the vertebral column is exposed and the lateral surface of the vertebral body is dissected to remove the soft tissue. A small awl is used to form a pilot hole for the main vertebral body screw. This pilot hole is usually close to the plane of the rib heads, parallel to the end plates in the axial plane, albeit perpendicular to the lateral vertebral body surface slightly directing anteriorly off the spinal canal. The length of the screw and screw tract preparation can be assessed based on previous MRI and/or CT based measurements of vertebral depth, using the surgeon's finger approaching the contralateral vertebral body wall or by assessment of vertebral depth at the discectomy level adjacent to the vertebra to be instrumented. The screw platform is then inserted in the desired orientation and screwed into place using a screw driver with the main vertebral body screw. Then the second smaller anchor screw is secured to the vertebra. Finally, the rods are locked to the screw-head platform. If there are multiple rods of different sizes, the larger diameter rod is secured first and then the smaller diameter rod.

(42) It will of course be understood that the present disclosure has been described above purely by way of example, and that modifications of detail can be made within the scope of the invention.

Claims

1. A spinal implant device comprising a first surface and a second surface that is opposite the first surface along a transverse direction, wherein the second surface is a bone facing surface, a first side and a second side that are opposite from one another along a longitudinal direction, perpendicular to the transverse direction; a first end, and a second end that is opposite the first end along a lateral direction, perpendicular to the transverse direction and the longitudinal direction, the spinal implant device defining: wherein the spinal implant device defines a main bore that extends along a central main bore axis that extends along the transverse direction, the main bore configured to receive a main vertebral body fixation element therethrough such that the main vertebral body fixation element is configured to extend into an underlying vertebra, wherein the spinal implant device defines an anchor bore that is offset from the main bore along the lateral direction and that extends along a central anchor bore axis, the anchor bore configured to receive an anchor fixation element therethrough, wherein the anchor bore does not pass through any rod receiving recesses, and wherein the first surface includes a convex portion that extends from the main bore and a concave portion that extends from the convex portion toward the second end.

2. The spinal implant device of claim 1, wherein at least a portion of the main bore is configured to receive a locking screw that secures a rod in a primary recess, the portion of the main bore having an upper surface and a lower surface, wherein the upper surface of the anchor bore is disposed below the lower surface of the portion of the main bore.

3. The spinal implant device of claim 1, wherein the spinal implant device has a first portion at the first end that defines a first height, and a second portion at the second end that defines a second height less than the first height.

4. The spinal implant device of claim 3, wherein the first height is measured from the second

surface to the first surface, and the second height is measured from the second surface to the first surface.

5. The spinal implant device of claim 4, wherein the first surface at the second portion is offset from the first surface at the first portion with respect to the transverse direction.

6. The spinal implant device of claim 4, wherein the second surface at the second portion is substantially aligned with the second surface at the first portion with respect to the transverse direction.

7. The spinal implant device of claim 3, wherein the first height is from a select plane to the first surface, and the second height from the select plane to the first surface, wherein the select plane is defined at the second surface and is perpendicular to the central main bore axis.

8. The spinal implant device of claim 3, wherein the main bore extends into the first portion and the anchor bore extends into the second portion.

9. The spinal implant device of claim 1, wherein a line that intersects the central main bore axis of the main anchor bore and the anchor bore axis of the anchor does not intersect any other screw holes.

10. The spinal implant device of claim 1, wherein the spinal implant device is devoid of any screw holes between the main bore and anchor bore.

11. The spinal implant device of claim 1, wherein the spinal implant device is devoid of screw holes between the main bore and the first side and between the anchor bore and the second side.

12. The spinal implant device of claim 1, wherein the anchor fixation element is a screw and the anchor bore is threaded to receive a thread on a head of the screw.

13. A system comprising the spinal implant device of claim 1 and the main vertebral body fixation element that is configured to be inserted through the main bore and into the underlying vertebra.

14. A system comprising the spinal implant device of claim 1 and the anchor fixation element, wherein the anchor fixation element is an anchor screw that is configured to be inserted through the anchor bore and into the underlying vertebra.

15. The system of claim 14, wherein the anchor bore has internal threading, and the anchor screw has a head that is threaded to engage the threading of the anchor bore.

16. The spinal implant device of claim 1, wherein the concave and convex portions of the first surface are between the main bore and the anchor bore.

17. The spinal implant device of claim 1, wherein the convex portion is a first convex portion and the first surface includes a second convex portion between the concave portion and the second end.

18. The spinal implant device of claim 17, wherein the main bore includes a main bore perimeter at the first surface and the anchor bore includes an anchor bore perimeter at the first surface, and wherein the entire anchor bore perimeter is offset below the main bore perimeter with respect to the transverse direction.

19. The spinal implant device of claim 1, further comprising a screw-head platform that includes a primary recess configured to receive a primary rod.
